

3-machine

Two maps, one extra +1.

Thesis (discrete): A bounded local kick can still run away if one residue is never paid back. Change that one kick and the orbit sits.

What is actually proved in Lean (no sorry)

`ThreeMachine.lean`

- `stepU n = if n%2=0 then n+3 else n+1` → `stepU (stepU n) = n+4` → `stepU^[2*k] n = n+4*k` → unbounded, steps ≤ 3
- `stepS n = if n%2=0 then n else n+1` → `stepS_lands_even` → `stepS^[k.succ] n = stepS n` → bounded after 1 tick

Current status

- `ThreeMachine.lean` : builds, 0 sorry - KEEPER
- `Main.lean` : replace with `Main_FIXED.lean` to close `diesel_principle` using `stepU` as witness
- `Stable.lean` : replace with `Stable_FIXED.lean` (no more `n = n` placeholder)

`lake build` will be 0 sorry after those two replacements.

No fluids, no mass, no time claims in Lean - those are notes in `UPT..._lean.txt`, not theorems.

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